

6.1. Automorphism groups admitting a subgroup of type V_4 . Except for those with automorphism group $C_2 \times C_5$, all families in the table above are specializations of the family of curves

$$C_{t,s} : y^2 = x^6 + tx^4 + sx^2 + 1$$

over a finite field $\overline{\mathbb{F}}_p$ with p odd. The quotient by the automorphism $(x, y) \mapsto (-x, y)$ produces the degree-two morphism

$$\phi : \begin{cases} C_{t,s} \longrightarrow E_{t,s} : v^2 = u^3 + tu^2 + su + 1 \\ (x, y) \longmapsto (u, v) = (x^2, y). \end{cases}$$

The complementary elliptic subcover of degree two is

$$\phi' : \begin{cases} C_{t,s} \longrightarrow E_{s,t} : v^2 = u^3 + su^2 + tu + 1 \\ (x, y) \longmapsto (u, v) = (1/x^2, y/x^3), \end{cases}$$

which can also be described as the quotient of $C_{t,s}$ by the automorphism $(x, y) \mapsto (-x, -y)$.

These two covers produce a $(2, 2)$ -isogeny

$$\Phi = \phi^* \times \phi'^* : \begin{cases} E_{t,s} \times E_{s,t} \longrightarrow \text{Jac}(C_{t,s}) \\ (P - \infty, Q - \infty) \longmapsto \sum_{R \in \phi^{-1}(P)} R - \sum_{R \in \phi^{-1}(\infty)} R + \sum_{R \in \phi'^{-1}(Q)} R - \sum_{R \in \phi'^{-1}(\infty)} R, \end{cases}$$

whose kernel is contained in $(E_{t,s} \times E_{s,t})[2]$. Write $u^3 + tu^2 + su + 1 = (u - \alpha_1)(u - \alpha_2)(u - \alpha_3)$. Then $u^3 + su^2 + tu + 1 = (u - \frac{1}{\alpha_1})(u - \frac{1}{\alpha_2})(u - \frac{1}{\alpha_3})$. Set $P_i^\pm = (\pm\sqrt{\alpha_i}, 0)$, so that $\phi^{-1}((\alpha_i, 0)) = P_i^+ - \infty + P_i^- - \infty$. Under the usual identification of E with its Jacobian variety, the kernel of Φ is

$$\{\infty \times \infty\} \cup \left\{ (\alpha_i, 0) \times \left(\frac{1}{\alpha_i}, 0 \right) : i \in \{1, 2, 3\} \right\}.$$

Indeed, we have $\Phi \left((\alpha_i, 0) \times \left(\frac{1}{\alpha_i}, 0 \right) \right) = \text{div}(x^2 - \alpha_i) = 0$.

Remark 6.3. *The elliptic curves $E_{t,s}$ and $E_{s,t}$ are in general not isogenous to each other.*

We can now use the results of Section 5.1 to determine the endomorphism ring of $\text{Jac}(C)$. Consider the dual $(2, 2)$ -isogeny $\widehat{\Phi} : \text{Jac}(C_{t,s}) \rightarrow E_{t,s} \times E_{s,t}$. It allows us to bound the endomorphism ring from above and below as follows

$$2 \text{End}(E_{t,s} \times E_{s,t}) \subset \text{End}(\text{Jac}(C_{t,s})) \subset \frac{1}{2} \text{End}(E_{t,s} \times E_{s,t})$$

where the inclusions are given by the maps:

$$\begin{cases} 2 \text{End}(E_{t,s} \times E_{s,t}) \longrightarrow \text{Jac}(C_{t,s}) \\ 2\psi \longmapsto \Phi \circ \psi \circ \widehat{\Phi} \end{cases} \quad \begin{cases} \text{End}(\text{Jac}(C_{t,s})) \longrightarrow \frac{1}{2} \text{End}(E_{t,s} \times E_{s,t}) \\ \varphi \longmapsto \frac{1}{2} \widehat{\Phi} \circ \varphi \circ \Phi \end{cases}$$

In order to finally identify $\text{End}(\text{Jac}(C_{t,s}))$ among orders which satisfy those bounds, we check which elements $\frac{1}{2}\psi \in \frac{1}{2} \text{End}(E_{t,s} \times E_{s,t})$ can be written as $\frac{1}{2}\widehat{\Phi} \circ \varphi \circ \Phi$, that is, when $\psi = \widehat{\Phi} \circ \varphi \circ \Phi$.

6.2. Automorphism type $C_2 \times C_5$. To conclude this section, we focus on the remaining case of the curve $C : y^2 = x^5 - 1$ defined over a finite field $\mathbb{F}_p$ with $p \neq 2, 5$.

We clearly have the inclusion $\mathbb{Z}[\zeta_5] \subset \text{End}(\text{Jac}(C))$. In the simple case, that is, when p is totally split in $\mathbb{Z}[\zeta_5]$, this is an equality. In the non-simple case, we can use the results of Section 5.1 since thanks to [72, Theorem 1.2] we do know the p -rank and the a -number of the Jacobian variety.